

3-Mark Questions & Answers — MPCA Module 6: Pentium 4 & ARM Processor

Q1. Compare 8086, 80386, Pentium 1, and Pentium 4 microprocessors.

Feature	8086	80386	Pentium 1	Pentium 4
Data Bus	16-bit	32-bit	64-bit	64-bit
Address Bus	20-bit (1 MB)	32-bit (4 GB)	32-bit (4 GB)	36-bit (64 GB)
Clock Speed	5 - 10 MHz	16 - 33 MHz	60 - 200 MHz	1.3 - 3.8 GHz
Pipeline	2-stage	3-stage	5-stage dual U/V	20-stage NetBurst

Q2. Explain ARM vs Thumb instruction states.

- **ARM State (32-bit):** Standard high-performance 32-bit instruction set with conditional execution on all instructions.
- **Thumb State (16-bit):** Compact 16-bit subset providing 30-40% code size reduction for memory-constrained embedded systems.